Abtin Aryanpour

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Education

University of Waterloo — Waterloo, ON
Candidate for BASc in Computer Engineering, Co-op Program — *Sep. 2026 – Apr. 2031*

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: PyTorch, NumPy, OpenCV, scikit-learn, Django, jQuery
Tools & Platforms: Git, GitHub Actions (CI), pytest, CMake, SQLite, AWS (EC2, CloudFront)
Concepts: Machine Learning, Computer Vision, Embedded Systems, Unit Testing, Web Development

Projects

Andon: Machine-Sound Anomaly Detection | *Python, C, Raspberry Pi Pico 2* — Oct. 2026 – Present

- Built a system that detects failing machines by sound, training a PyTorch CNN only on healthy audio; it reached 92.13% mean AUC on held-out DCASE 2020 data vs. 79.70% for the DCASE autoencoder baseline (3 seeds)
- Validated the evaluation by reproducing the DCASE baseline within 0.88 AUC points on all 6 machine types
- Developed a streaming green/yellow/red alert using control-chart (SPC) limits, catching 65.57% of 30 s faults with a 10.57 s mean delay at 5.53 false alarms/hour on held-out clips
- Quantized the CNN to int8 for a Cortex-M33 microcontroller with a NumPy fixed-point reference, shrinking it to 98.7 KB (from 383.0 KB) at 92.08% AUC; exported C headers and test vectors to check the C port bit for bit in CI

CyberShield: Full-Stack Discussion Forum | *Python, Django, JavaScript, Tailwind CSS* — Mar. 2025

- Built a full-stack forum as the only developer on a 3-person team using Django, jQuery (AJAX) and SQLite, winning 1st of 32 teams at CNLC 2025
- Implemented voting, sorting, filtering and fuzzy search (Fuse.js) through JSON endpoints, with no page reloads
- Secured accounts with Django auth, CSRF protection, reCAPTCHA and moderator roles; deployed on AWS

Experience

Autonomy Software Member — Sep. 2026 – Present
Waterloo Aerial Robotics Group — *Waterloo, ON*

- Sole developer of the team's groundside image stitcher in Python (PyTorch, OpenCV, XFeat, RANSAC), merging out-of-order, GPS-tagged photos from 3 simulated drones into one georeferenced map; 80 tests
- Fused image matches with GPS for heading, aligning photos to 4.3–6.9 cm on average at 3 m simulated GPS error
- Designed a bundle-adjustment-style least-squares solver, cutting simulated misalignment from 13–41 to 4–7 cm
- Hardened matching against 180-degree turns and repetitive terrain, forming one map on 10/10 simulated runs
- Added lens undistortion and blur, exposure and tilt checks through 2 review rounds on PR #215 (in review); the blur check sorted 2,730 simulated photos under 9 noise and blur conditions with 0 errors

Treasurer — Sep. 2025 – Jun. 2026
Bayview Secondary School — *Richmond Hill, ON*

- Managed a $21,000 budget for 42 clubs serving 2,000 students, approving each allocation with club leaders

Homework Helper & Assistant Teacher — Sep. 2022 – Jul. 2025
Spirit of Math Schools Inc. — *Richmond Hill, ON*

- Ran a drop-in homework classroom as one of two staff (2024–2025), helping about 30 students per session
- Taught 1–2 classes a week of up to 16 students (2022–2024), covering lessons and contest-style algebra

Awards & Achievements

1st Place (2025) and 4th Place (2024): Coding & Programming, Canadian National Leadership Conference
Euclid (86/100), Hypatia (33/40) Honour Roll; Fermat, CSMC Distinction: Waterloo Math Contests, 2025–26
Excellence in Mathematics: Bayview Secondary School Mathematics Department (2025, 2026)